Friction Before Failure: Making Review Burden Visible in Regulated Document Work

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Abstract Deviation records and Corrective and Preventive Action (CAPA) documents in pharmaceutical manufacturing environments are reviewed by multiple actors across iterative revision cycles and formal approval chains — yet the coordination work that produces delay remains invisible to any single participant. This paper argues that review friction is not only a workflow inefficiency but an HCI problem about how interaction conditions shape what coordination work becomes visible and actionable before failure occurs. I introduce a judgment-preserving observability layer that surfaces hidden coordination work without automating quality decisions. I report a two-part prototype-based study with $N = 20$ QA/QC-experienced participants: a six-tab workflow dashboard that surfaces revision burden, comment distribution, and document journey patterns as interpretable signals, and a TF-IDF pattern detection layer that identifies recurring issue clusters across review cycles. Neither component modifies validated documentation systems. Participants consistently rated both components highly and expressed adoption intent, while reporting that friction visibility enabled them to interpret delay as a structural condition rather than an individual failure. The study's central empirical finding is a dissociation between immediate legibility and final endorsement: the pre-review checklist was immediately accessible because it addresses a problem practitioners already recognize, while the pattern detection layer required initial conceptual orientation because it addresses a problem practitioners have not previously had the tools to name — structural friction accumulation. Both arrived at equivalent ratings. I introduce the judgment-preservation constraint as a novel design boundary for workflow support research in regulated environments, and argue that making hidden coordination work legible is a precondition for more equitable accountability in regulated sociotechnical organizations.

Keywords: review friction, workflow visibility, CAPA, pharmaceutical manufacturing, GMP, invisible work, judgment-preserving design, regulated environments

Introduction

In pharmaceutical manufacturing environments governed by Good Manufacturing Practice (GMP) regulations, deviation and corrective action (CAPA) records are among the most consequential documents a quality organization produces. Their review involves multiple actors —

quality assurance specialists, subject matter experts, department managers — cycling through iterative rounds of comment, revision, and approval. These review processes are routinely experienced as slow. But the standard diagnosis of that slowness — too many reviewers, unclear ownership, insufficient training — misidentifies the problem. The core issue is not delay itself. It is that the coordination work producing the delay is structurally invisible.

Review friction accumulates gradually: a wording comment triggers a revision cycle; a reviewer reopens a document closed in a previous round; a pattern of minor but recurring feedback generates disproportionate turnaround time across a document class. These patterns are real, consequential, and structural — but they remain distributed across individual comment threads, document versions, and reviewer roles. No single actor has a complete view of them. As a result, friction becomes visible only after delays have already escalated, and accountability is assigned retroactively, without access to the coordination conditions that produced the outcome. This is not only a workflow problem. It is an interaction problem: the system does not make coordination work visible at the moment it could be interpreted and acted upon.

This paper introduces a design intervention that addresses this observability problem directly: an external workflow visibility layer that surfaces review friction as interpretable signal without modifying validated documentation systems or automating quality decisions. The distinction matters: in GMP environments, human interpretive authority over quality decisions is not merely a design preference but a regulatory requirement. The design challenge is therefore to make friction visible while leaving judgment intact — the judgment-preservation constraint that defines the core design boundary of this work.

I report a two-part prototype-based study. In the first part, I developed a six-tab workflow dashboard over a synthetic dataset of 100 deviation-CAPA review records, and evaluated it with $N = 20$ QA/QC-experienced participants. In the second part, I added a TF-IDF pattern detection layer to surface recurring issue patterns across review cycles. Neither component modifies validated documentation systems; both operate as external observability layers.

The contributions of this work are fourfold. First, I introduce review friction visibility as a design problem specific to regulated document environments, identifying the judgment-preservation constraint as a novel design boundary for workflow support research. Second, I demonstrate that a lightweight observability layer can surface hidden coordination work without automating quality decisions. Third, I identify a dissociation between immediate legibility and final endorsement as a design finding: tools that address structurally invisible problems will routinely require practitioners to first acquire the conceptual vocabulary to name those problems, and this interpretive overhead is a design responsibility rather than a user limitation. Fourth, I offer a theoretical framing that connects HCI concerns about workflow visibility to broader questions about how accountability is structured in regulated sociotechnical organizations.

Related Work

Awareness and Coordination in Collaborative Work Systems

A foundational concern in CSCW research is how participants in distributed collaborative work maintain awareness of each other's activities. Dourish & Bellotti (1992) establish that awareness — understanding of the activities of others that provides context for one's own action — is critical to successful collaboration and is poorly served by active, explicit information-sharing mechanisms. Their study of shared editing systems demonstrates that passively available awareness information, embedded in the shared workspace itself, allows participants to coordinate dynamically without the overhead of explicit communication. Applied to regulated document review, this framework implies that the coordination work occurring across review cycles — who has commented, which rounds are open, where revision burden is accumulating — should ideally be available as ambient, interpretable signal rather than requiring active compilation by individual reviewers. The observability layer in this study is designed to produce precisely this: awareness information embedded in the workflow itself, requiring no active compilation from participants who currently must construct a partial picture of coordination patterns from individual threads and document states.

Invisible Work and the Limits of Formal Indicators

Coordination work in distributed systems often becomes invisible precisely because it is effective: when practitioners manage revision loops, absorb comment burden, and route documents through informal channels, the effort expended does not appear in any formal record. Star & Strauss (1999) theorize this as invisible work — labor that is performed but not captured by organizational indicators, and therefore not attributed to the structural conditions that produce it. In regulated review environments, this invisibility has specific implications: the coordination work that generates delay sits outside CAPA records, audit trails, and quality metrics. When a CAPA document requires three revision cycles because a cross-reference issue recurs across rounds, no quality metric captures this pattern; the audit trail records only the final closure date. The observability layer developed in this study makes this invisible work legible — not by automating its interpretation, but by surfacing the patterns that indicate its presence. The goal is not to eliminate the work but to attribute it correctly: to the structural conditions that produce it rather than to the individual practitioners who absorb it.

Situated Action and the Conditions of Review Work

Suchman's (1987) account of situated action establishes that action is not plan-driven but shaped by the immediate conditions of its execution — what is observable, reachable, and interpretable

at the moment a decision must be made. Applied to document review, this implies that the quality of review decisions is shaped not only by reviewer expertise but by the conditions of visibility under which those decisions are made. A reviewer who cannot see how a document has moved through prior rounds, what patterns of comment have accumulated, or where the current cycle sits relative to historical revision burden is making decisions in conditions of reduced observability. Suchman's framework grounds the study's central argument: making coordination conditions visible is not a secondary support function but a precondition for the quality of review work itself. When reviewers act under conditions of partial visibility, they produce outcomes that reflect those conditions — outcomes that the quality system then attributes to reviewer judgment rather than to the observability conditions that shaped it.

Articulation Work and the Distribution of Coordination Labor

Schmidt & Bannon (1992) introduce the concept of articulation work to describe the coordinative labor that makes other work possible — the work of scheduling, allocating, routing, and monitoring that sustains collaborative activity. In regulated review workflows, articulation work takes the form of tracking revision states, compiling comment patterns, identifying responsible parties for recurring issues, and managing the informal communication channels through which review progress is communicated. This work is performed, but it is not attributed to the coordination structure that requires it. The observability layer in this study addresses articulation work at the structural level: by surfacing revision burden, comment distribution, and pattern accumulation as shared representations, it moves articulation labor from individual, invisible effort to collective, visible infrastructure.

Human-AI Interaction and Compliance Constraints

The pattern detection layer in this study uses TF-IDF similarity clustering to surface recurring issue patterns across review cycles. The design of this layer is governed by a principle that Amershi et al. (2019) formalize across guidelines for human-AI interaction: AI systems should make clear what they can and cannot do, display contextually relevant information, and support human control rather than supplant it. In regulated manufacturing environments, this principle is not merely a design preference but a compliance requirement: quality decisions must be made and documented by qualified human reviewers, and any system that automates or pre-empts those decisions introduces a regulatory risk. The pattern detection layer's design response to this constraint is specific: it surfaces clusters and scores as observability outputs that describe what has happened, not recommendations about what should happen next. The interface makes this constraint explicit through design: outputs are labeled as pattern summaries rather than recommendations, and the system's limitations are stated in the interface itself.

Prototype-Based Evaluation

The methodological approach in this study builds on established HCI practice for evaluating prototype systems. Nielsen (1994) demonstrates that think-aloud evaluation with small participant samples reliably surfaces usability problems and can be combined with post-task assessment to understand both navigational behavior and perceived utility. Hutchinson et al. (2003) establish the logic of using a prototype simultaneously as a design intervention and as an analytic probe — a technology probe makes visible the conditions of use it is designed to address, rather than simply testing whether a given interface is usable. This dual function is central to the present study: the dashboard and pattern detection layer were not deployed to measure their own effectiveness in isolation but to make visible the structural conditions of review friction that conventional workflow analysis cannot observe.

AI-Assisted Decision Making and Judgment-Preserving Design

Within HCI, work on AI-assisted decision making has established that interface structure actively shapes which information becomes salient to decision makers. Wang & Yin (2021) demonstrate that explanation interfaces alter how decision makers engage with outputs, with effects contingent on domain expertise — findings that establish observability tools as active shapers of interpretation rather than neutral displays. Green & Chen (2019) show that algorithm-in-the-loop settings systematically constrain the conditions under which independent judgment is possible, establishing that the structural conditions of human-algorithm interaction determine whether human interpretive authority is genuinely preserved. In regulated environments where judgment preservation is a compliance requirement rather than a design preference, these findings ground the observability layer's design logic: surfacing coordination patterns without recommending actions preserves the interpretive authority that GMP regulation requires, and instantiates the judgment-preservation constraint as a design boundary that distinguishes this work from conventional workflow automation.

Prior work has not examined review friction visibility as a design problem in regulated document environments, where the judgment-preservation constraint uniquely restricts what computational support is permissible. This study addresses that gap directly.

Method

Study Context

This study was conducted in the context of pharmaceutical manufacturing quality systems operating under GMP regulations. Deviation and CAPA review workflows in these environments

involve multiple reviewers, iterative revision cycles, and formal approval chains — all of which generate coordination work that is rarely visible as a unified pattern to any individual participant. The pharmaceutical GMP context was selected not as a convenience sample but as a domain in which the conditions the study addresses — coordination invisibility, judgment-preservation constraint, and accountability consequences — are particularly pronounced and formally codified.

Dataset Construction

The study used a synthetic dataset of 100 deviation-CAPA review records generated programmatically in Python. Because actual facility records are subject to non-disclosure obligations, the dataset was constructed to reflect the structural and terminological characteristics of real GMP review workflows — including comment categories, review round sequences, severity classifications, turnaround times, document reopening events, and reviewer role assignments — without reproducing proprietary content. The judgment-preservation constraint applies equally to synthetic and live records: observability tools must operate externally over data rather than within the document management system. Synthetic generation from structural parameters is the standard approach for research in regulated manufacturing contexts where live data access is unavailable, and the structural validity of the findings does not depend on the specific content of the records but on the patterns of coordination and friction they generate.

Phase 1: Workflow Dashboard Prototype

Based on the synthetic dataset, I developed a six-tab workflow dashboard as a Streamlit web application (Figures 1–6). The dashboard was designed as an external observability layer — it reads from the dataset but does not modify, version, or interact with any validated documentation system. The six tabs surface distinct dimensions of review burden, each addressing a specific component of the coordination invisibility identified in the problem framing.

- **Overview** (Figure 1): comment category distribution, severity breakdown, average turnaround by reviewer role, and review round distribution across the synthetic dataset.
- **Heatmap** (Figure 2): revision bottleneck visualization across reviewer–document type combinations, with revision loop frequency indicators.
- **Checklist** (Figure 3): a pre-review scaffold with a rule-based comment classifier to support consistent review preparation, and a comment category $\times$ average turnaround chart.
- **Document Journey** (Figure 4): full review lifecycle trace with round-by-round summary for individual records.

- **Before/After** (Figure 5): comparative workflow support visualization showing anticipated changes across key dimensions.
- **Reviewer Simulation** (Figure 6): comment framing tool with friction estimation to support reflection on review patterns.

The design of each tab was governed by the judgment-preservation principle: the dashboard surfaces patterns and distributions as interpretable signals, but does not recommend actions, assign responsibility, or produce outputs that function as quality decisions. This constraint was operationalized specifically: the Overview tab shows distributions but draws no conclusions about their causes; the Heatmap shows role-level patterns but does not identify responsible individuals; the Checklist surfaces preparation gaps but does not evaluate reviewer performance; the Document Journey shows lifecycle states but does not recommend actions; the Reviewer Simulation shows historical patterns but frames them as reflection support rather than performance assessment.

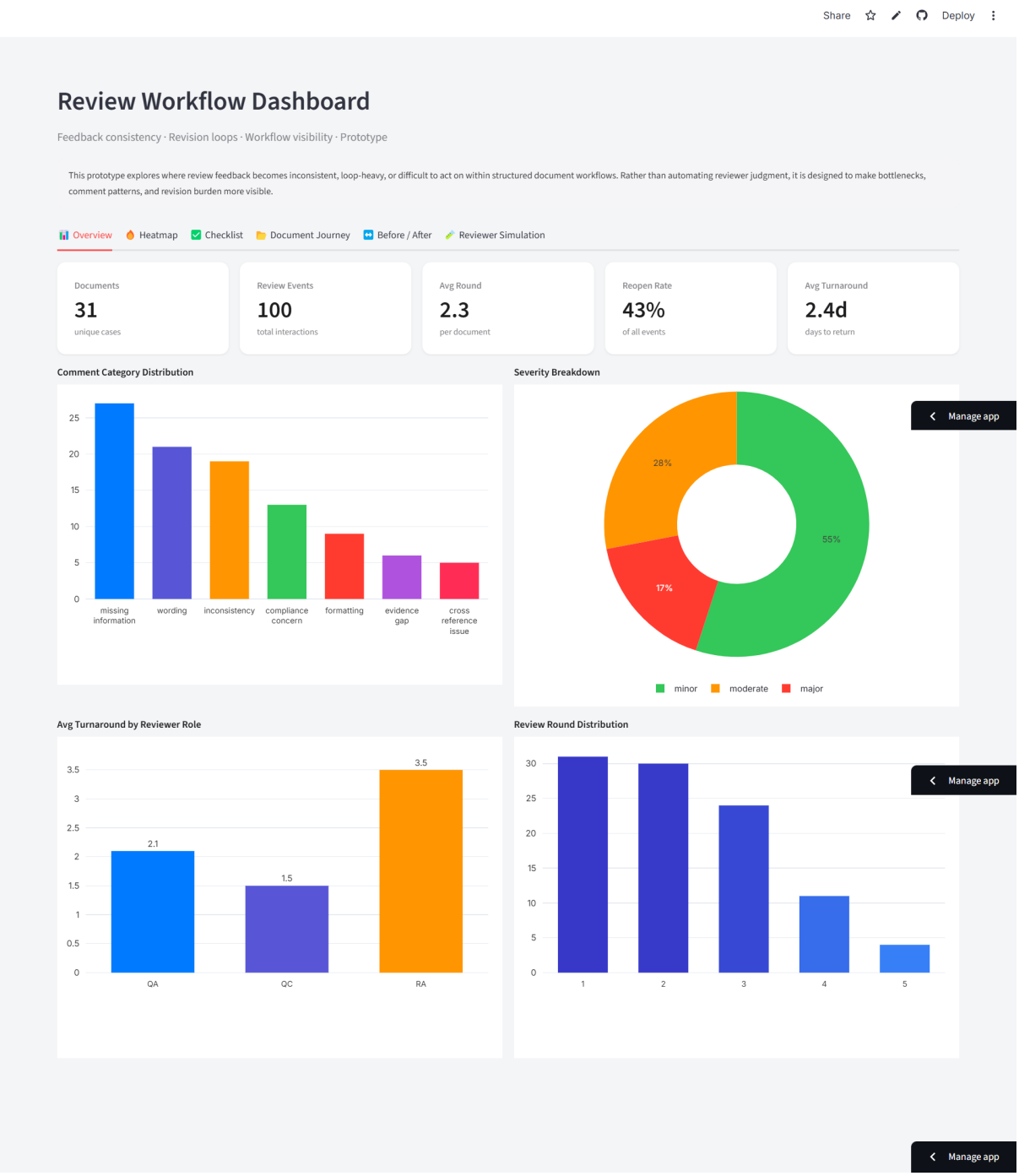


Figure 1: Overview tab of the workflow dashboard, showing comment category distribution across the 100-record synthetic dataset, severity breakdown (55% minor, 28% moderate, 17% major), average turnaround by reviewer role (QA: 2.1d, QC: 1.5d, RA: 3.5d), and review round distribution. The RA role's substantially higher turnaround is visible as a pattern.

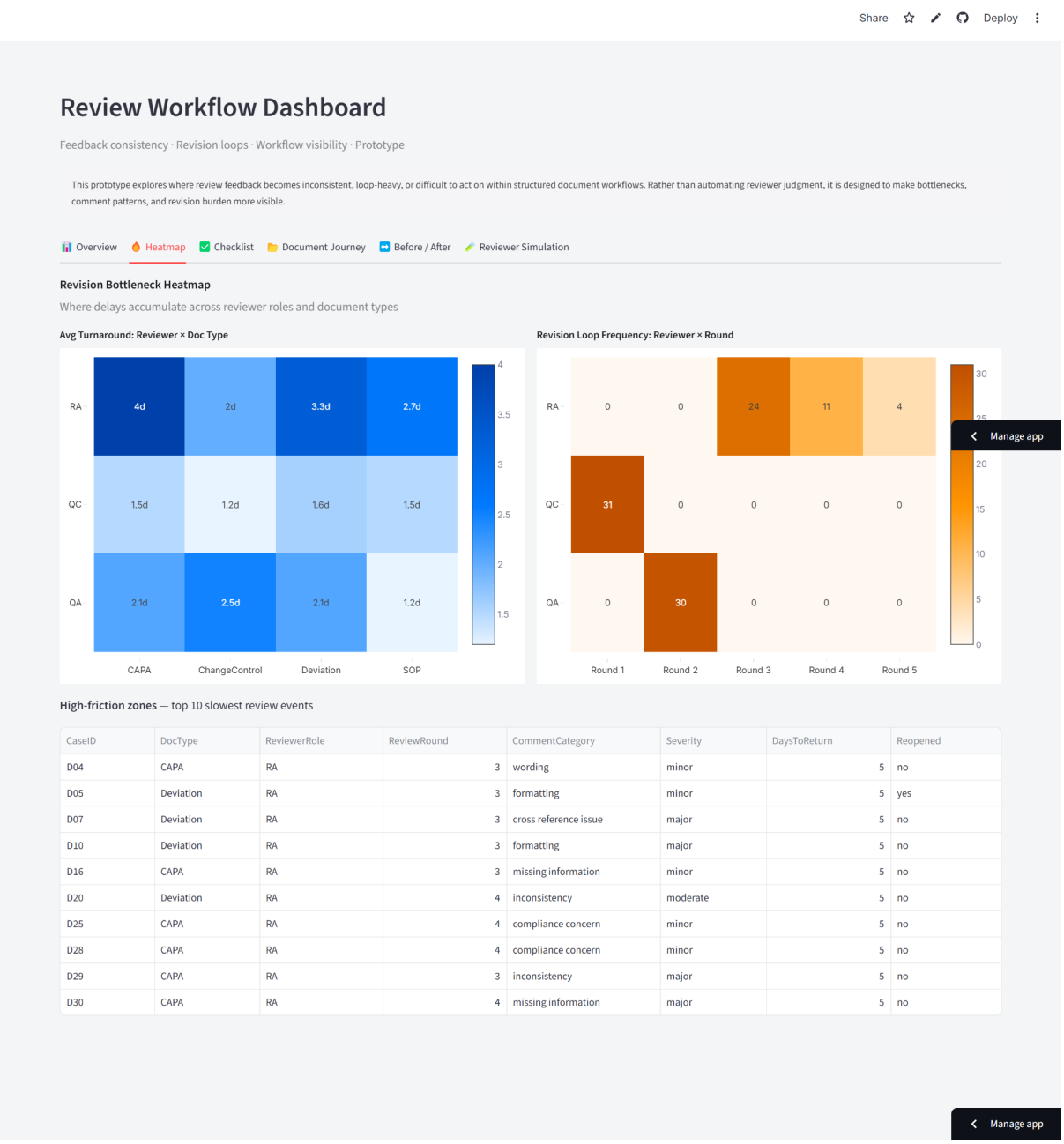


Figure 2: Heatmap tab, showing average turnaround by reviewer role and document type (left) and revision loop frequency by reviewer and round (right). QC loops concentrate in Round 1 (31 events), QA loops in Round 2 (30 events), and RA loops spread across Rounds 3–5 — a structural pattern indicating that different reviewer roles generate friction at systematically different stages of the review lifecycle.

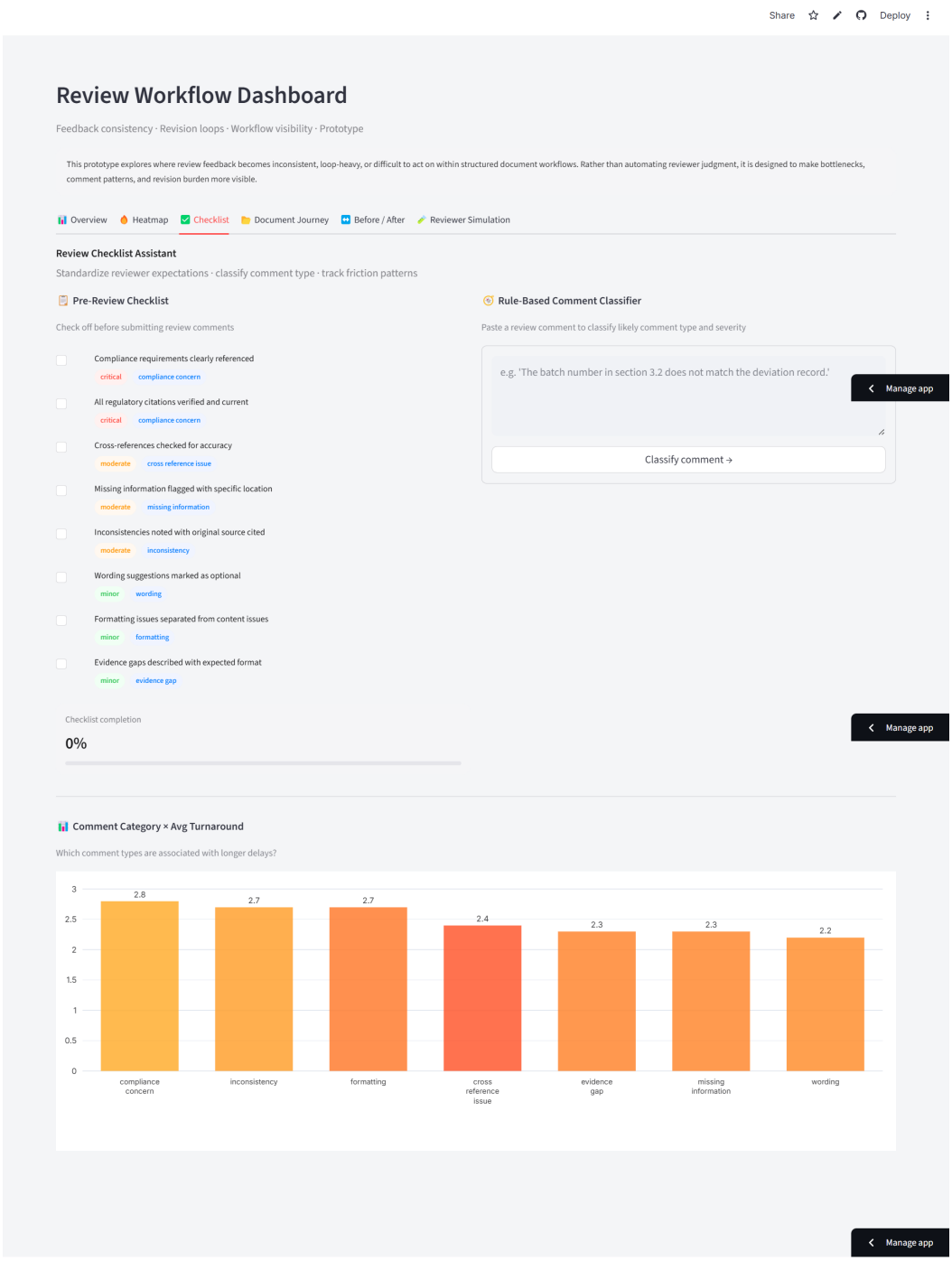


Figure 3: Checklist tab, showing the pre-review checklist scaffold (left) and rule-based comment classifier (right). The checklist was the most immediately legible component in the pilot evaluation — rated 5.0/5 without requiring conceptual reorientation.

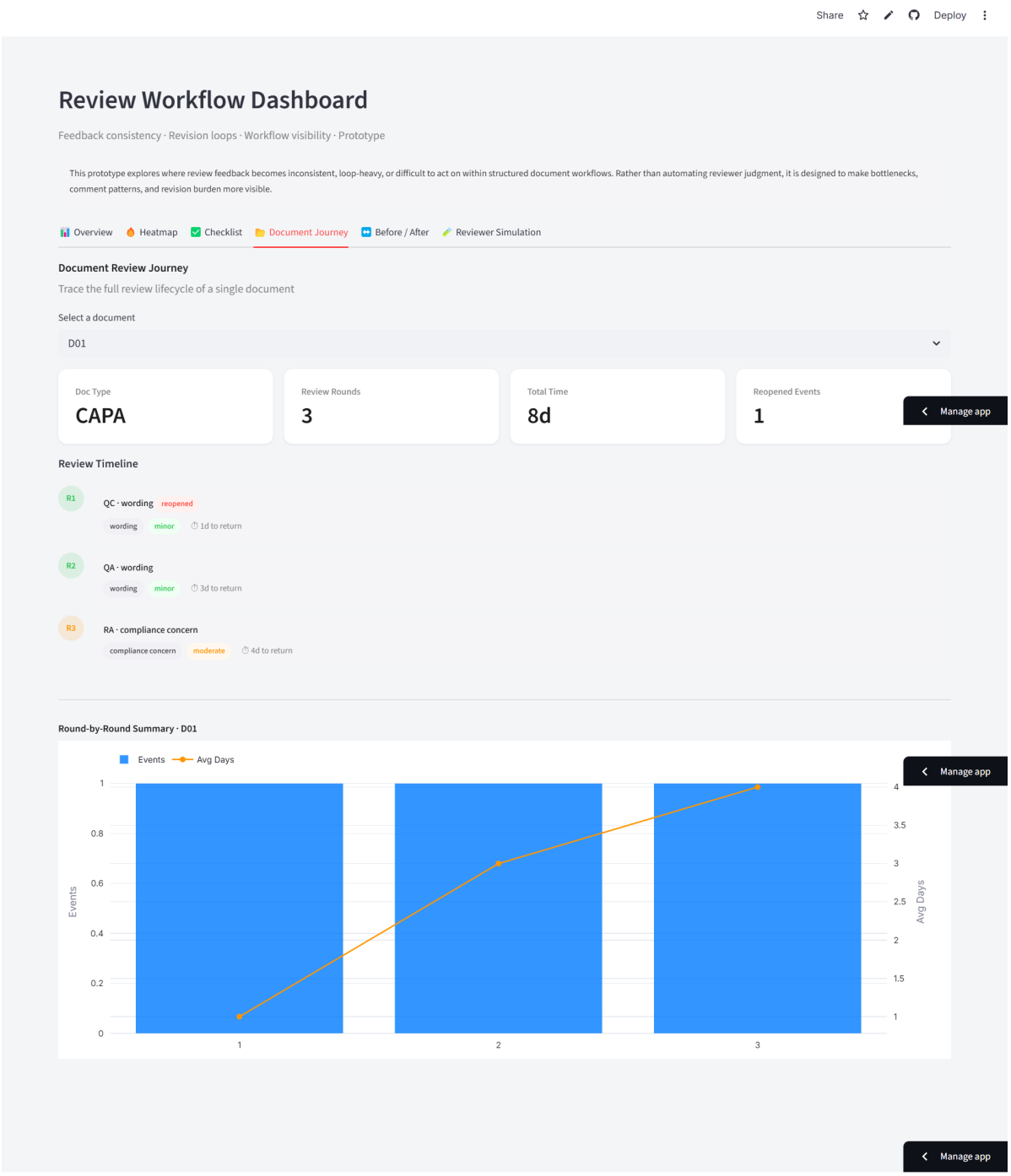


Figure 4: Document Journey tab, showing the full review lifecycle of a single document (D01: CAPA, 3 review rounds, 8 days total, 1 reopened event). The review timeline shows round-by-round activity with comment category, severity, and return time for each event.

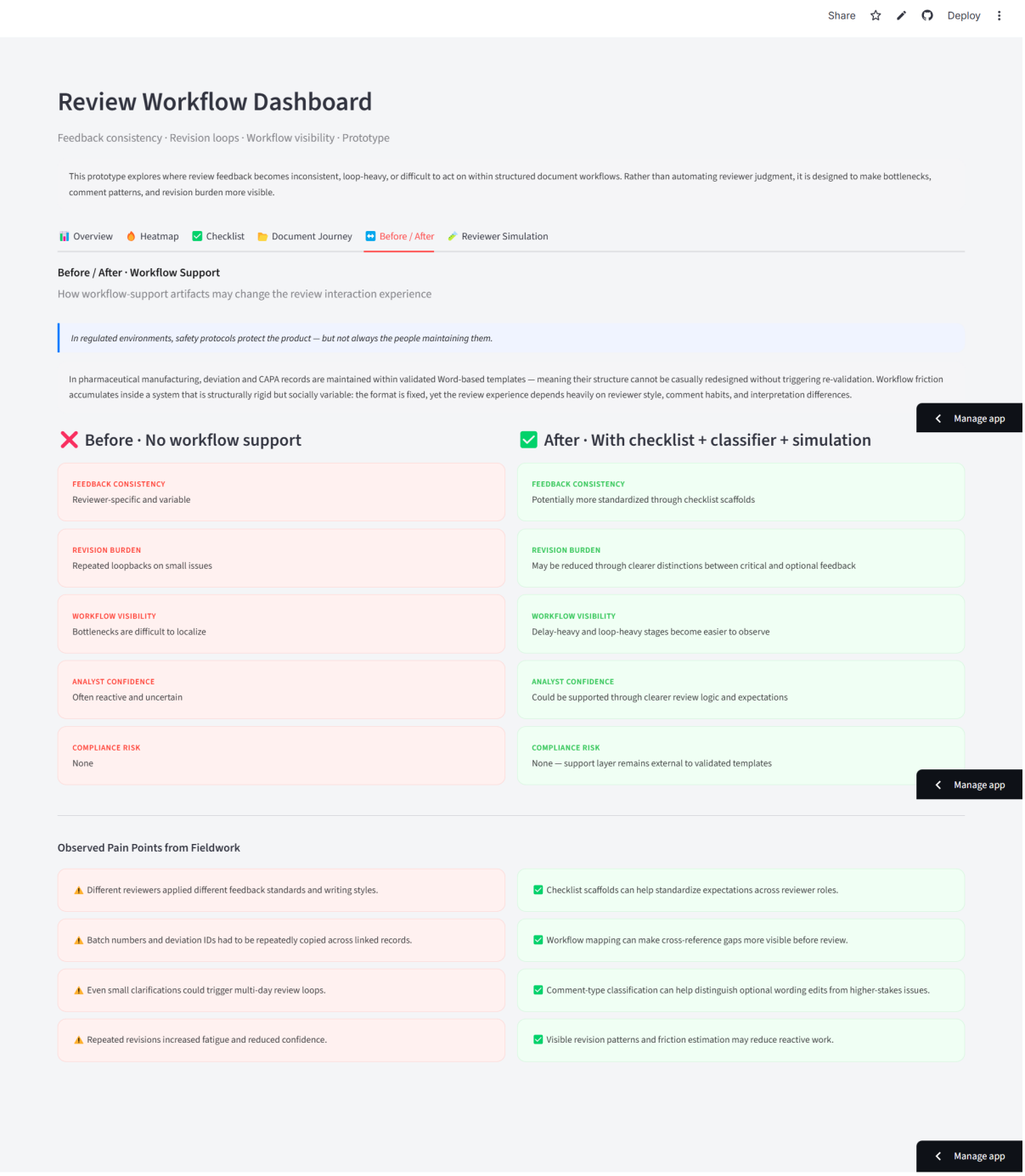


Figure 5: Before/After tab, showing anticipated changes across five dimensions (feedback consistency, revision burden, workflow visibility, analyst confidence, compliance risk) when the observability layer is applied.

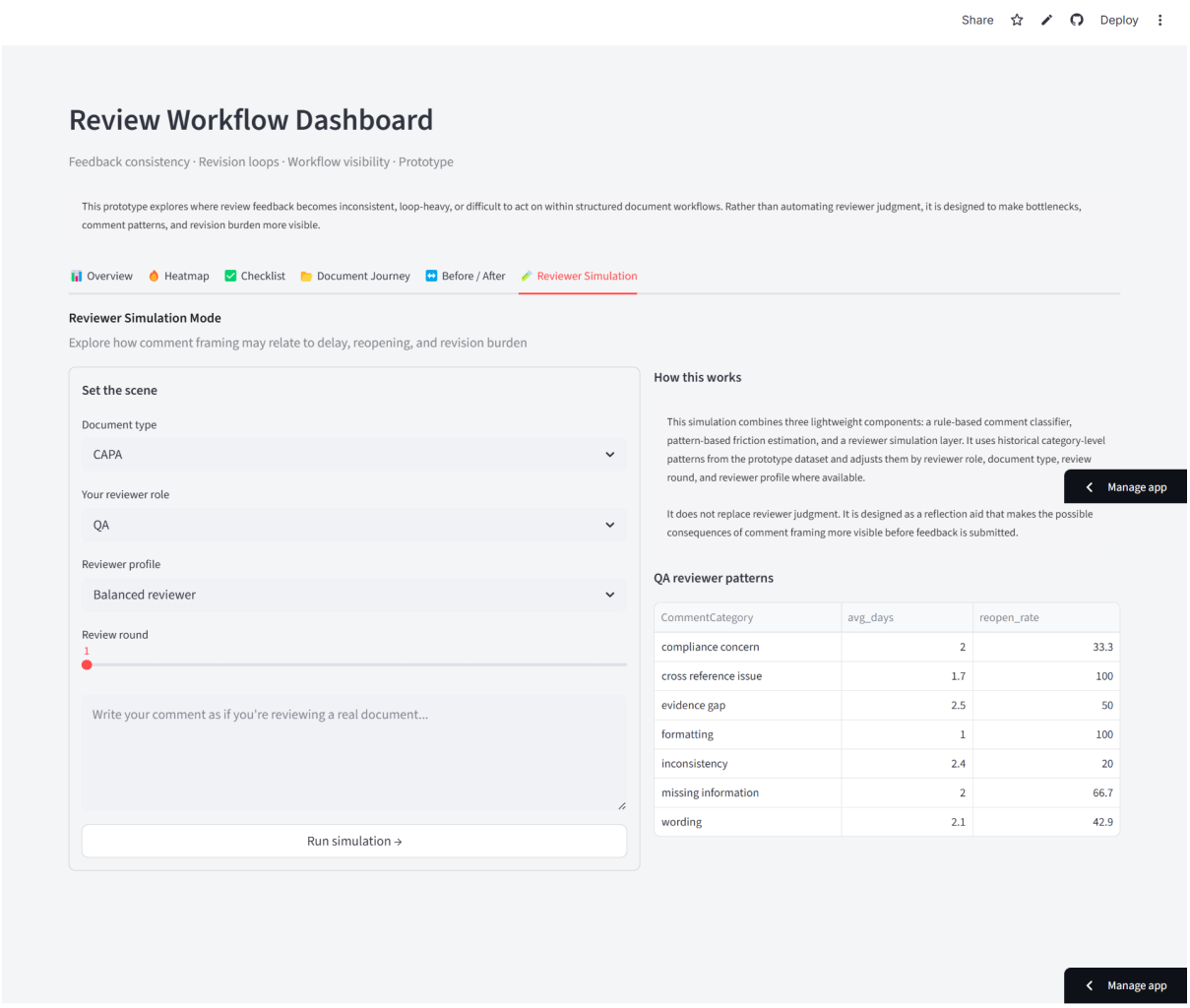


Figure 6: Reviewer Simulation tab, showing the comment framing tool with document type, reviewer role, reviewer profile, and review round controls. The QA reviewer patterns table (right) surfaces historical category-level patterns — for example, cross-reference issues are associated with a 100% reopen rate, while inconsistency comments average 2.4 days and a 20% reopen rate.

Phase 2: Pattern Detection Layer

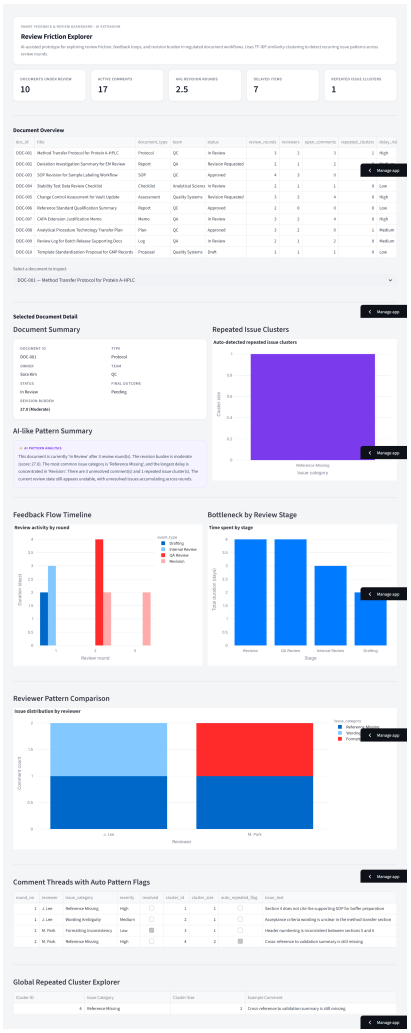


Figure 7: Review Friction Explorer: the pattern detection layer adding TF-IDF clustering over the dashboard dataset. For a selected document (DOC-001), the layer surfaces: a document summary with revision burden score (27.0, moderate); auto-detected repeated issue clusters (Reference Missing dominates); a plain-language pattern summary describing the document's review state; a feedback flow timeline by round; bottleneck by review stage; reviewer pattern comparison; and comment threads with auto pattern flags.

The workflow dashboard addressed friction visibility at the aggregate level — surfacing patterns across documents and reviewer roles. It did not address the per-document question: for a specific document currently under review, what structural patterns have accumulated, and at which stage has friction concentrated? The TF-IDF pattern detection layer (Figure 7) addresses this residual gap, applying the same judgment-preservation principle to document-level pattern analysis.

The layer is organized around three functions. **TF-IDF clustering** detects recurring patterns across comment text without manual tagging — grouping similar comments across documents

and reviewers into issue clusters that surface the structural categories of friction accumulation. **Revision burden scoring** quantifies the coordination load associated with individual records, combining review round count, comment volume, reopen events, and cluster concentration into a single interpretable score. **Pattern summaries** describe each document's review state in plain language — identifying the dominant issue category, the stage at which bottlenecks have concentrated, and whether the current review state is stable or accumulating. These summaries are observability outputs, not recommendations: they describe what has happened, not what should happen next.

Throughout, the layer makes explicit what it does not do: it does not recommend actions, make accountability decisions, replace reviewer judgment, or modify validated documentation systems (Amershi et al., 2019).

Pilot Evaluation

I evaluated both components with $N = 20$ participants recruited from a pharmaceutical quality team with QA/QC review experience. The same-sample design eliminated between-participant variance in domain knowledge as a confound; recruiting from a single expert team ensured consistent baseline familiarity with GMP review conventions, though it may have inflated acceptance ratings relative to a more diverse sample. The evaluation was conducted outside any live validated system — a necessary condition in GMP contexts where introducing an unvalidated interface into a production quality workflow would itself constitute a compliance risk.

Participants completed task-based evaluation scenarios using both the dashboard and the pattern detection layer, providing feedback on purpose clarity, friction visibility, checklist usability, simulation interpretability, and adoption intent. Post-task discussion probed the conceptual accessibility of the friction-visibility framing, with particular attention to the dissociation between components that address problems practitioners already recognize (the checklist) and components that address problems practitioners have not previously had the tools to name (the pattern detection layer).

Findings

Friction Visibility as Fairness Infrastructure

The most consequential participant response was not about interface usability but about what visibility of review burden would change organizationally. All 20 participants anticipated reduced accountability asymmetry from the dashboard — a finding that emerged consistently across both submitter and reviewer roles without prompting. Participants described a recurring

condition in current practice: review delays accumulate, documents circulate without clear progress signals, and when closure finally fails or slows, responsibility is assigned without access to the coordination conditions that produced the outcome.

Two patterns in the qualitative data were consistent across participants. First, accountability was routinely attributed to the most recently visible actor: “the delay gets attributed to whoever submitted the document, not to the review structure that produced it.” Second, the invisibility of coordination conditions was experienced as organizationally consequential, not merely inconvenient: “this would make it harder to blame the person at the end of the queue when the problem was in the middle of the process.” A third participant described the asymmetry from the reviewer’s perspective: “I know when I’m adding to the burden, but I can’t see the full picture. I don’t know if my comment is the fifth time someone has flagged the same issue or the first.”

What this response makes visible is a shift in the unit of accountability. When coordination patterns were not visible, delay was described in outcome terms — who failed to close, who took too long, who kept reopening. When friction patterns were surfaced, the same delays were reinterpreted as structural accumulations across the workflow: patterns that were predictable, attributable to design conditions, and therefore addressable without blame assignment. In this sense, friction visibility functioned not only as workflow support but as fairness infrastructure — a mechanism for making the conditions of accountability visible before outcomes are assigned.

Role-Differentiated Utility

Participants described the dashboard’s value differently depending on their role in the review process. Document submitters primarily valued visibility into where their document stood and what type of review activity was generating delay: “I can see my CAPA has been in Round 3 for eight days, but I can’t see why. This would show me.” QA reviewers valued the dashboard differently: it offered a workload monitoring function that current systems do not support, making it possible to regulate their own review burden across documents rather than responding to each in isolation: “I have fourteen documents in review right now. I have no idea which ones are about to escalate. This would change that.”

This role differentiation is significant for deployment: the dashboard is not a tool for one side of a coordination asymmetry but a shared infrastructure that makes the asymmetry visible to all participants simultaneously. All 20 participants expressed adoption intent regardless of role, and the role-differentiated framing of utility suggests that the value of observability is not zero-sum — making friction visible benefits submitters and reviewers in different but compatible ways.

The Checklist vs. Pattern Detection Dissociation

The TF-IDF clustering layer received a final usability rating of 5.0/5 and was endorsed by all 20 participants as useful once demonstrated — yet it required initial orientation to interpret in a way that the checklist did not. The checklist was immediately accessible because it addresses a problem practitioners already recognize: inconsistent review preparation produces predictable downstream friction, and a structured preparation scaffold addresses a burden practitioners can name. The pattern detection layer required initial orientation because it addresses a problem practitioners have not previously had the tools to name — structural friction accumulation as a measurable, per-document property.

This initial unfamiliarity did not produce rejection. Once the framing was explained, all 20 participants endorsed the concept as genuinely useful. One participant described the reorientation explicitly: "I never thought of revision loops as something you could measure. But once you see it that way, it's obvious that some documents are just structurally harder." A second participant described the shift in problem framing: "I always assumed the hard documents were hard because of the content. This shows me they're hard because of the review pattern. That's a completely different problem." A third described the adoption barrier directly: "the tool makes sense. The question is whether the organization wants to see this."

This dissociation — between immediate legibility and final endorsement — is the study's central empirical finding. Usefulness and immediate recognizability are not the same thing. A tool can be highly useful for a problem that practitioners have not yet learned to see. The appropriate metric for evaluating accountability-critical design interventions is not whether practitioners immediately recognize their value, but whether they endorse them once the problem has been made legible. The interpretive overhead of problem-naming is a design responsibility, not a user limitation.

The Organizational Adoption Problem

All 20 participants expressed adoption intent; all 20 simultaneously identified a structural barrier to deployment. The barrier was not technical feasibility or user acceptance — participants were clear that they wanted the system. The barrier was organizational: whether regulated organizations would adopt a system that makes review reasoning more transparent.

The specific form of the concern was consistent across participants. Making friction patterns visible creates records of coordination conditions that were present and not addressed. In a regulatory environment subject to audit, these records may surface accountability conditions that current practice leaves invisible. "Right now, if a CAPA takes three revision cycles because the same cross-reference issue keeps coming up, there's no record of that pattern. This would create one." A second participant described the organizational calculus more directly: "the system is

useful because it makes patterns visible. It's also risky for the same reason. You can't unsee what you've documented."

This combination — unanimous adoption intent alongside unanimous identification of an organizational barrier — is not contradictory. It reflects the gap between individual preference and organizational incentive. Individual practitioners want the conditions the dashboard creates: better visibility into coordination patterns, more equitable attribution of delay, reduced informal burden. Organizations may not want the transparency the dashboard produces, because that transparency makes visible choices and patterns that are currently made and accumulated invisibly. Whether regulated organizations can tolerate — or are required by their regulatory environment to produce — that level of transparency is a question that interface design cannot answer. It requires engagement with quality culture, audit standards, and the regulatory frameworks that govern what review records must contain.

Discussion

Observability as Interaction Design

The central contribution of this work is a reframing of workflow support in regulated environments. The standard framing treats workflow support as optimization: the goal is to make review processes faster, more consistent, and less prone to error. Regulatory HCI proposes a different goal: making the conditions of review work visible, so that the conditions that shape accountability can be understood, discussed, and redesigned rather than absorbed individually.

When coordination work is invisible, participants act under conditions of partial visibility and attribute outcomes based on incomplete information. By making coordination patterns visible, the observability layer reshapes how workflow outcomes are interpreted without automating those outcomes. This aligns with Suchman's (1987) account of situated action: action is shaped by what is visible and interpretable at the moment it is taken. Applied to document review, this means that accountability is shaped not only by what reviewers decide, but by what the system makes available for those decisions to be understood against. The practitioner who closes a CAPA without addressing a recurring cross-reference issue may not be negligent; they may be acting under conditions in which the pattern was not visible. Making the pattern visible changes what actions are possible, what interpretations are available, and what accountability can be assigned — not by automating any of these, but by creating the observability conditions under which they can occur.

Invisible Work and the Organizational Stakes of Legibility

The role-differentiated responses documented in the findings reflect the asymmetric distribution of invisible work that Star & Strauss (1999) describe. Document submitters absorb the cost of review delays without visibility into their causes; QA reviewers absorb coordination burden without visibility into its cumulative distribution across documents. Both forms of invisible work are real, consequential, and structurally produced — yet neither is captured by any quality metric or attributed to the coordination design that produces it.

The dashboard addresses both forms of invisibility by making coordination patterns a shared, interpretable representation. That all 20 participants across both roles reported adoption intent suggests that the observability layer is not a tool for one side of a coordination asymmetry but a shared infrastructure for making that asymmetry discussable. The organizational stakes of this legibility are real: a system that makes coordination patterns visible also makes visible the choices and patterns that those conditions enabled to remain unexamined. The question is not whether making invisible work visible is valuable — participants were unanimous that it is — but whether organizations are structured to tolerate the accountability implications of that visibility.

Judgment-Preserving Design in Compliance-Constrained Environments

The design constraint that governed both components — that computational outputs must support human judgment rather than substitute for it — is not merely a normative preference. In GMP environments, quality decisions must be made and documented by qualified human reviewers; any system that automates or pre-empts those decisions introduces a compliance risk. The pattern detection layer's approach — surfacing TF-IDF clusters and burden scores as observability signals rather than recommendations — is one instantiation of the judgment-preservation principle (Amershi et al., 2019).

The finding that clustering outputs required initial interpretive orientation, despite ultimately receiving 5.0/5 ratings, points to a productive tension: the more a system preserves human judgment by withholding recommendations, the more it requires users to bring their own interpretive frame to the output. This tension is not resolvable by making observability outputs more directive — that would violate the judgment-preservation constraint. It is addressable by building the interpretive bridge explicitly, as the checklist does: establishing the problem category before presenting the tool that addresses it. The checklist is not merely a simpler entry point; it is the instrument through which practitioners develop the conceptual vocabulary that makes the pattern detection layer's value legible.

The Checklist as Interpretive Bridge

The checklist's immediate legibility relative to the pattern detection layer suggests a practical adoption pathway and a broader design principle. For adoption: checklist-first deployment establishes the dashboard's credibility within existing review practice, followed by progressive introduction of pattern detection outputs as users develop the interpretive frame to use them. The conceptual barrier to friction visibility is real but surmountable — all 20 participants endorsed the concept once demonstrated.

The broader principle is this: in accountability-critical environments, tools that address structurally invisible problems will routinely require practitioners to first acquire the conceptual vocabulary to name those problems. This is not a usability failure but a feature of the design challenge. Making structural conditions visible requires building a new problem category, not just a new interface. The appropriate design response is not to reduce the tool's ambition to match existing problem categories, but to build the interpretive infrastructure — simpler entry points, problem framing, vocabulary — that allows practitioners to develop the categories the tool requires. The checklist functions as precisely this kind of interpretive bridge.

Limitations

Several limitations bound the interpretation of these findings. The synthetic dataset underrepresents the terminological and structural variance of live review records, which may have reduced the discriminability of clustering outputs and contributed to the high acceptance ratings. Participants were recruited from a single team, limiting generalizability. The study does not establish effects on formal quality metrics — CAPA closure time, deviation recurrence rates, or audit findings. Future work should deploy both components in live review environments, examine the organizational conditions under which friction visibility is adopted or resisted, and investigate whether making coordination patterns visible changes the distribution of accountability outcomes over time.

Conclusion

This paper introduced review friction visibility as a design problem in regulated document environments, and demonstrated that an external observability layer can surface hidden coordination work without modifying validated documentation systems or automating quality decisions. Observability does not eliminate friction; it makes friction legible as an interaction condition that shapes accountability before failure occurs. All 20 participants rated both components 5.0/5, expressed adoption intent, and anticipated reduced accountability asymmetry from friction visibility.

The study's central empirical finding is a dissociation between immediate legibility and final endorsement: the checklist was immediately accessible because it addresses a problem practitioners already recognize, while the pattern detection layer required initial conceptual orientation because it addresses a problem practitioners have not previously had the tools to name. Both arrived at equivalent ratings. The interpretive overhead of the pattern detection layer is not a usability failure — it is a signal that making structural friction visible requires building a new problem category, not just a new interface.

Four contributions follow. First, review friction visibility is introduced as a design problem specific to regulated document environments, where the judgment-preservation constraint uniquely restricts what computational support is permissible. Second, the external observability layer is demonstrated as a viable design strategy for surfacing coordination patterns within compliance boundaries. Third, the dissociation between immediate legibility and final endorsement is identified as a design finding with implications for how accountability-critical interventions should be evaluated: endorsement after problem framing, not immediate recognition, is the appropriate criterion. Fourth, the organizational adoption problem — the gap between individual adoption intent and organizational deployment feasibility — is identified as a structural finding about the accountability assumptions embedded in existing compliance systems, requiring organizational and regulatory engagement that extends beyond interface design.

References

Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*.

Dourish, P., & Bellotti, V. (1992). Awareness and coordination in shared workspaces. In *Proceedings of the 1992 ACM Conference on Computer-Supported Cooperative Work* (pp. 107–114).

Green, B., & Chen, Y. (2019). The principles and limits of algorithm-in-the-loop decision making. *Proceedings of the ACM on Human-Computer Interaction, 3*(CSCW), Article 50.

Hutchinson, H., Mackay, W., Westerlund, B., Bederson, B. B., Druin, A., Plaisant, C., Beaudouin-Lafon, M., Conversy, S., Evans, H., Hansen, H., Roussel, N., & Eiderbäck, B. (2003). Technology probes: Inspiring design for and with families. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (pp. 17–24).

Nielsen, J. (1994). *Usability engineering*. Morgan Kaufmann.

Schmidt, K., & Bannon, L. (1992). Taking CSCW seriously: Supporting articulation work. *Computer Supported Cooperative Work, 1*(1–2), 7–40.

Star, S. L., & Strauss, A. (1999). Layers of silence, arenas of voice: The ecology of visible and invisible work. *Computer Supported Cooperative Work, 8*(1–2), 9–30.

Suchman, L. A. (1987). *Plans and situated actions: The problem of human-machine communication*. Cambridge University Press.

Wang, X., & Yin, M. (2021). Are explanations helpful? A comparative study of the effects of explanations in AI-assisted decision-making. In *Proceedings of the 26th International Conference on Intelligent User Interfaces* (pp. 318–328).